

DATA 205 Data Science Final Project Plan

Unemployment Rates in the U.S.

Ria Rhodes

Data:

For my primary source of data, I will be using the U.S. Bureau of Labor Statistics data on unemployment. Specifically I will be looking at their Local Area Unemployment Statistics and Labor Force Statistics including the National Unemployment Rate ([BLS Unemployment Data](#)). This is a collection of unemployment data from across the United States of America, detailing percentage rates across different years. These datasets will be used for the brunt of my statistical analysis and visualization work, although I will also utilize the [U.S. Census Bureau Population Data](#) for investigation into trends and population densities. Although there are no known issues, the only potential complication I can assume will arise regarding these datasets would be with dates syncing up. Due to a variety of different worldly events, there may be gaps in which data was not added to these datasets, leading to complications regarding dataset synchronization.

Research Goals:

My primary goal with this research is to understand unemployment rates with relevancy to historical events or social stigmas at the time which may have affected them. Because of this I plan on doing a lot of work with the unemployment rate in counties (thus the supplemental datasets), looking into local news stories and business monopolies for finer details that may affect employment.

Primary Questions:

- a. What trends have affected unemployment rates over time?
- b. What historical events have affected unemployment rates over time?

For both of these questions, they can be analyzed by developing a graph denoting change in unemployment rate over time, then doing extensive background research on the political climate, supreme court cases, and social stigmas to put any notable portions of the graph into context.

For thorough analysis, this should be completed with the overarching dataset initially to gauge the presidency, global issues, or biases at the times of notable outliers. If anything is exceptionally within relation to a certain location, look further into that (For example, if there was a large forest fire up the west coast near a time of notable unemployment, look deeper into counties within California, map out rates of unemployment on a GIS map, and investigate into property damages or death counts via the Census API within remarkable areas).

Secondary Questions:

- a. Which counties per state have the highest unemployment rate?
This can be achieved by taking the average unemployment rate per county over time (Using the supplemental datasets only) and comparing the

- numbers. For a visual aid, GIS maps could be created with varying sizes of markers to denote different rates of unemployment.
- b. Are there inherent linking factors in the living conditions that develop differing rates of unemployment across the country?
Begin by doing research on what the different housing densities (City v. suburbs v. sparse urban) may be in counties with notable unemployment rates (lowest rate, highest rate, etc.) as per what was discovered in the GIS map. Look for any similarities or correlations.
 - c. Is there an association between population densities and unemployment rates?
Calculate population density by taking the population count as per the Census API and divide it by the physical size of the county. Compute correlation tests, spearman and pearson, to find associations between the unemployment rates and the newly created population density variable.
 - d. With our modern sociopolitical climate, what can we expect out of the future regarding unemployment?
Consider trends noted in previous questions, research into our current economic state, and develop an assumption based on modern-day global and political changes to decipher what we can assume the future of unemployment will be like.
 - e. Is there a notable difference in unemployment rates across different geographic areas of the U.S.? (East coast v. west coast, south v. north, etc)
Look at the previous GIS map for a visual, looking for differences in markings on the map. Statistically, split the states into quadrants and find the averages in unemployment per area over time. See if certain areas have notably higher/lower unemployment rates or if there are any odd irregularities.
 - f. Are there patterns in historical increases in unemployment? Can they be avoidable?
Look at the list of historical reasonings for unemployment I have developed. Are any of them similar in content or context? Do any of them follow a similar train of thought? Has there been any passed legislation changing this problem? Collecting the answers to these questions, I can decipher if there are notable triggers to unemployment and ponder on what can be done to avoid these complications.

Risks and Mitigations:

Of the things that could go wrong, I am by far worried about the quality of my data. It was quite the struggle to even find any data applicable to this data, and I am manually sorting through very raw data pulled straight from the government. At most, what I can do is triple check my work and corroborate some of the more general results with my research. I cannot imagine there will be actual complications with the production of my analysis and research, as they are quite straightforward processes and just require time and elaboration.

Milestones:

Now that my data is 110% finalized and committed to, here is a timeline of what I believe my process will be.

February 20th, found initial dataset

March 3rd, initial visualizations and data exploration

March 7th, primary questions found

March 12th, all supplemental data found

March 18th, initial cleaning and reformatting data done

March 20th, secondary questions found

March 26th, choose to also use and learn Census API

April 2nd, create and polish faceted graph of unemployment over time

April 16th, create and polish full GIS map for all counties across all states

April 30th, have all statistical analysis and calculations complete

May 3rd, clean R markdown and work on written report

May 10th, have slides created and finalized, practice and memorize material

May 14th, final presentation and report